

Moon Formation by Triple Phase Transition in the Differentiating Proto-Earth

*Bingham-Herschel rheology, oblique instability,
Orbital Seeding Body and progressive dynamo*

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Abstract. The origin of the Moon remains one of the major enigmas of planetary cosmogony. This work proposes a mechanism based on the physics of threshold complex systems: the formation of the Moon results from a *triple phase transition* within the fully molten proto-Earth undergoing progressive differentiation.

The first transition is **rheological**: the silicate magma obeys a Bingham-Herschel (BH) law with a yield stress τ_y , which governs the cohesion of the intertropical Coherent Magmatic Torus (TMC, from the French *Tore Magmatique Cohérent*). This TMC is of 'Aā nature: its biphasic structure — a mobile plastic core and a fragmented clinker crust — constitutes an intrinsic thermal insulator that maintains the torus coherence over the entire active window (≈ 60 Myr), without recourse to an external blanketing atmosphere. The second transition is **mechanical**: kinetic energy accumulates in the TMC — a toroidal structure emerging in the oblique latitude band $|\varphi| < 30^\circ$ — until crossing a critical hypersonic ejection threshold ($Ma \approx 11.6$), triggered by Mathieu resonance with the proto-lunar tide. The third transition is **magnetic**: the terrestrial dynamo emerges progressively as Fe-Ni segregation toward the growing core surpasses the dynamo convection threshold, becoming detectable at ≈ 4.2 Ga through the magnetization of Hadean zircons (Tarduno et al., 2025).

These three transitions are governed by a single driver: the progressive differentiation of iron toward the core. This driver simultaneously controls the rheological evolution of the TMC, the energy available for episodic ejections, and the growth of the core that powers the dynamo. The T-Tauri Sun, as an actively coupled actor, maintains the fully molten state, modulates the effective Bingham threshold τ_y through its radiative flux, and provides impulsive perturbations via coronal mass ejections (CMEs) that can trigger the first ejection episode. The temporal coincidence between the T-Tauri phase (~ 4.55 – 4.50 Ga) and the proposed ejection window (4.55–4.49 Ga) is **causal, not accidental**.

The capture of the ejected sheets by the Orbital Seeding Body (CAO, from the French *Corps d'Amorçage Orbital*) is modeled as a process of **non-ballistic plastic accretion**: the cohesive Bingham-Herschel sheet loses its energy by viscoplastic dissipation within the sphere of influence of the proto-satellite, not by elastic collision of independent particles. The capture fraction f_{cap} is expressed by a closed analytical law coupling interception geometry and plastic capture number Π_{cap} .

This episodic accretion produces a **concentric internal lunar architecture in successive layers**: each sheet carries the geochemical imprint of the Earth's mantle at the precise moment of its ejection, generating a radial Fe/Si gradient decreasing from the interior outward. The current state of lunar seismological data is compatible with this stratification, without confirming or refuting it. The ongoing and forthcoming seismological missions — Farside Seismic Suite (FSS), Lunar Environment Monitoring Station (LEMS, Artemis III), and the international network under construction — offer the most direct observational framework to test this prediction, prior to and independently of any 3D numerical simulation.

The nominal rotation period is $T_{\text{rot}} = 3.5$ h, consistent with planetary accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010). The stability coefficient λ is derived and discussed (Section 6.2).

Posture note. This work proposes a theory. All hypotheses are explicit. All limitations are acknowledged. The observational and numerical validation programme is defined. Final validation belongs to forthcoming lunar seismological data and, in a second stage, to 3D SPH (Smoothed Particle Hydrodynamics) simulations.

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List of abbreviations and main symbols

Symbol / Abbr.	Definition
TMC	Coherent Magmatic Torus (from French <i>Tore Magmatique Cohérent</i>): toroidal structure emerging in $ \varphi < 30^\circ$, of 'A $\bar{a}$ ' rheological type
BH	Bingham-Herschel law: yield-stress rheology
'A$\bar{a}$'	Morphological lava type: biphasic structure (plastic core + fragmented clinker crust) constituting an intrinsic thermal insulator
SPH	<i>Smoothed Particle Hydrodynamics</i>
TTP	Triple Phase Transition
TP	Taylor-Proudman constraint
CAO	Orbital Seeding Body (from French <i>Corps d'Amorçage Orbital</i>): first solid-plastic object placed in orbit during the first ejection episode
CME	<i>Coronal Mass Ejection</i> from the T-Tauri Sun
OML	Lunar Magma Ocean
FSS	<i>Farside Seismic Suite</i> : first seismometer on the farside of the Moon (NASA/CLPS, launch 2025–2026)
LEMS	<i>Lunar Environment Monitoring Station</i> : portable seismometer deployed by Artemis III astronauts
LVZ	<i>Low Velocity Zone</i> in the deep lunar mantle, enriched in ilmenite
Hf-W	Hafnium-tungsten isotopic system: chronometer of planetary differentiation
ATR	Thermoremanent magnetization
Ga	Giga-annum (billions of years before present)
ε	Axial obliquity [degrees]. Nominal range $[40^\circ, 70^\circ]$
Ω	Angular velocity of the proto-Earth $[\text{rad s}^{-1}]$. $\Omega = 2\pi/T_{\text{rot}} \approx 4.987 \times 10^{-4} \text{ rad/s}$ for $T_{\text{rot}} = 3.5 \text{ h}$
Ro	Rossby number $= U/(2\Omega L)$. $\text{Ro} \ll 1$: geostrophic regime. $\text{Ro}_{\text{TMC}} \approx 1.57 \times 10^{-4}$
Ma	Mach number $= U/c_s$. $\text{Ma} \approx 11.6$ during hypersonic ejection
Λ_{Els}	Elsasser number $= \sigma B^2/(\rho\Omega)$: ratio of magnetic to Coriolis forces
L_β	Rhines scale $= \sqrt{U_{\text{turb}}/\beta} \approx 0.52 R_\oplus$
β	Meridional gradient of the Coriolis parameter $= 2\Omega \cos \varepsilon / R_\oplus$
U_{crit}	Critical ejection velocity $= \sqrt{2 g_{\text{eff}} R_\oplus} \approx 10.41 \text{ km/s}$
g_{eff}	Effective gravity $= g_0 - \Omega^2 R_\oplus \sin \varepsilon$
$f_{\text{rad}}, f_{\text{horiz}}$	Radial and horizontal components of the Coriolis force
α	Exponent of the radial velocity profile: $\partial U / \partial r \approx -\alpha U / r$. Conjecture: $\alpha > 1$ (super-rigid regime)
λ	Radial stability coefficient. $\lambda > 0$: instability (ejection); $\lambda < 0$: stability (confinement)

Symbol / Abbr.	Definition
U_1, U_2	Boundaries of the instability window for $\alpha > 1$
τ_y	BH yield stress [Pa]. Depends on T (Arrhenius law) and T-Tauri flux
J_2	Dynamic oblateness coefficient of the proto-Earth
N	Number of ejection episodes. Triangulated to $\approx 7 \pm 2$
M_{ej}	Ejected mass per episode $\approx M_{\text{Lune}}/N$
M_{Lune}	Lunar mass $\approx 7.34 \times 10^{22}$ kg
R_{Roche}	Roche radius $\approx 3.6 R_{\oplus}$
τ_{cd}	Recharge time of the TMC between two ejection episodes
η_{TMC}	Conversion efficiency of rotational energy $\rightarrow$ TMC
Π	Cohesion parameter $= \rho U^2 / \tau_y$
f_{cap}	Orbital capture fraction (non-ballistic plastic accretion)

1 Introduction

Posture note

This work proposes a theory. The formulations used deliberately reflect this posture: the theory proposes, describes, predicts — it does not state absolute certainties. All hypotheses are explicit. All limitations are acknowledged. The validation programme is defined, with a clear hierarchy: lunar seismological observation first, 3D SPH simulation as an indispensable but later complement.

1.1 The three observational constraints

The origin of the Moon is one of the unresolved questions of modern planetology. A recent review of geophysical and geochemical constraints concludes that there is currently “no unambiguous geochemical or isotopic evidence for the role of an impactor in the formation of the Moon”; that the Moon would be a devolatilized fragment of the Earth’s mantle after core formation; and that the giant impact remains “the worst available model, except for all the others” (Sossi et al., 2025). This work proposes a physically motivated alternative, based on the internal dynamics of the differentiating proto-Earth.

Three observational constraints are particularly discriminating for any formation mechanism.

Isotopic similarity. The quasi-identical isotopic composition of the Earth and Moon for oxygen, chromium and titanium, established at less than 5 ppm (Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025), indicates that the Moon is made of the same material as the Earth’s mantle. This constraint is satisfied *mechanically* by the proposed mechanism, whose source material is the Hadean mantle contemporary with each ejection.

Iron depletion. The Moon is significantly depleted in iron relative to the Earth’s mantle, pointing to a source material from a mantle already undergoing siderophile differentiation. This gradient is here a *direct consequence* of the unique driver of the theory: progressive Fe-Ni segregation.

Angular momentum. The angular momentum budget of the Earth-Moon system requires a proto-Earth in very rapid rotation ($T_{\text{rot}} \approx 3\text{--}5\text{ h}$) at the time of formation (Čuk & Stewart, 2012; Canup, 2012). The adopted value, $T_{\text{rot}} = 3.5\text{ h}$, is consistent with planetary accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010) and constitutes a dynamic constraint on the ejection mechanism (Section 5).

The proposed mechanism satisfies these three constraints not by *a posteriori* adjustment but as direct consequences of its internal dynamics.

1.2 Why a flux mechanism and not a particle one

The majority of lunar formation models treat the material as particles or ballistic projectiles. This representation is ill-suited to the proto-Earth of 4.55 Ga ago: its entire accessible volume is a cohesive fluid with a rheological threshold, in rapid rotation, fully molten. Such a fluid develops large-scale coherent structures (zonal jets, planetary waves, toroidal tori) that have no equivalent in granular or ballistic mechanics.

Physical note

Structured continuous flux versus particle approach. The system remains within a continuous description (velocity field, density, rheological stresses), even if there is local rupture or shear. Coherence is macroscopic, not microscopic. The relevant distinction is not between flux and spray, but between **structured continuous flux** (strong internal connectivity) and **dislocated continuous flux** (reduced connectivity). In both cases, the SPH description remains valid. The Coherent Magmatic Torus (TMC, defined in Section 4) may undergo internal fragmentation without ceasing to be a coherent flux at large scales.

1.3 Architecture of the triple transition and unique driver**Key result**

This work proposes that the formation of the Moon is the by-product of a triple phase transition (TTP) in the differentiating proto-Earth. **The rheological and mechanical transitions** couple in real time: Fe-Ni differentiation governs simultaneously the rheology of the magma and the energy available for ejections. **The magnetic transition** is a delayed consequence of the same process, emerging ≈ 300 Myr after the extinction of ejections. The term “triple phase transition” designates a **unique driver** — iron differentiation — that produces three ruptures of physical regime on distinct timescales.

1.4 Explicit hypotheses

This work rests on the following hypotheses, all explicit:

1. **Total melting.** The proto-Earth is entirely molten from the surface down to the deep mantle layers undergoing segregation. No stable proto-crust can be maintained during the active window (4.55–4.49 Ga).
2. **Chaotic obliquity.** The axial obliquity ε varies chaotically in $[40^\circ, 70^\circ]$ in the absence of a stabilizing satellite (Laskar et al., 1993a; Laskar & Robutel, 1993b).
3. **Super-rigid profile (conjecture).** The radial velocity profile of the TMC satisfies $\alpha > 1$ (super-rigid regime) — **central conjecture, not demonstrated to date**, to be validated by 3D SPH simulation. Section 4.5 discusses the status of this hypothesis and the robustness of the mechanism for $\alpha \leq 1$.
4. **Progressive differentiation.** Fe-Ni differentiation is continuous and progressive over ≈ 60 Myr, constrained by hafnium-tungsten chronology (Hf-W, Kleine et al. 2002).
5. **Nominal rotation.** The rotation period is $T_{\text{rot}} = 3.5$ h, consistent with planetary accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010).

2 Initial state of the proto-Earth**2.1 A self-gravitating fluid sphere: total melting**

The proto-Earth at 4.55 Ga is a self-gravitating fluid sphere, entirely molten from the surface down to the deep mantle layers undergoing segregation. This is not a “magma ocean” superficial layer resting on a solid substrate: it is a single volume of material with a rheological

threshold, subjected to rapid rotation and its own gravity, without a stable internal solid interface during the active window.

Four mechanisms maintain this state of permanent melting: the gravitational energy of accretion ($\sim 2.5 \times 10^{32}$ J), the decay of short-lived radioactive isotopes (^{26}Al , ^{60}Fe), the continuous bombardment by planetesimals (Elkins-Tanton et al., 2011), and the heat of segregation of the forming Fe-Ni core.

The impossibility of a stable proto-crust results from five concomitant processes: fracturing by impactors; Taylor-Proudman (TP) columns arising from rapid rotation; vigorous convection; a surface temperature $> 3\,000\text{--}4\,000$ K maintained by the T-Tauri Sun (Feigelson & Montmerle, 1999); and a silicate atmosphere at $2\,000\text{--}4\,000$ K preventing any lasting surface cooling.

Table 2: Proposed internal structure of the proto-Earth ($t = 0\text{--}60$ Myr after Solar System formation).

Envelope	State	Density (kg m^{-3})	Temperature (K)	Dynamic role
Fe-Ni core	Liquid, forming	$\sim 10\,000$	$> 4\,000$	Active segregation (Hf-W) — driver of the TTP
Silicate mantle	Convecting	$\sim 4\,500$	$3\,000\text{--}4\,000$	Thermal reservoir
External magmatic mass	BH yield-stress fluid	$\sim 3\,300$	$2\,000\text{--}3\,500$	TMC source
Proto-atmosphere	Dense silicate gas	~ 16	$2\,000\text{--}4\,000$	Confinement + feedback + T-Tauri erosion

2.2 Progressive Fe-Ni differentiation: unique driver and Hf-W constraint

Gravitational segregation of iron and nickel toward the forming core is a continuous process over ≈ 60 Myr. Droplets of liquid Fe-Ni, being denser ($\rho \approx 10\,000 \text{ kg m}^{-3}$), percolate through the magmatic mantle and accumulate at the centre. This segregation is constrained by hafnium-tungsten chronology (Hf-W), which places core-mantle differentiation between ≈ 30 and ≈ 50 Myr after Solar System formation (Kleine et al., 2002).

Key result

Progressive Fe-Ni differentiation is the common driver of the three phase transitions. It continuously depletes the mantle in siderophiles — mechanically explaining the lunar iron deficit and producing the concentric lunar stratigraphy (Prediction P6, Section 13). It fuels core growth, triggering the dynamo with a ≈ 300 Myr delay (Section 11). It modifies the rheology of the external magmatic mass (TMC source) via temperature and composition.

2.3 Bingham-Herschel rheology: first phase transition

The constitutive material of the magmatic mass is not a classical Newtonian fluid. It is a **yield-stress fluid**, governed by the Bingham-Herschel (BH) law:

$$\tau = \tau_y + k \cdot \dot{\gamma}^n \quad \text{for } \tau > \tau_y, \quad (1)$$

where τ_y is the yield stress [Pa], k the consistency [Pa s^n], $\dot{\gamma}$ the shear rate [s^{-1}], and n the behaviour index. Below τ_y , the material behaves as an elastic solid; above, it flows in a coherent manner.

The value of τ_y for a silicate magma at 3 000–3 500 K is estimated between 10^2 and 10^4 Pa (Giordano et al., 2008). Dissolved water and CO_2 lower this threshold by a factor of 2 to 10 depending on volatile content (Giordano et al., 2008), which is favourable to ejection. The behavioural exponent $n = 1.2$ (shear-thickening fluid) is retained for a partially crystallized magma with a solid fraction $\phi \approx 0.1$ – 0.3 , consistent with the work of Caricchi et al. (2007).

The threshold temperature below which τ_y becomes prohibitive for ejection ($\tau_y \gtrsim 10^6$ Pa) is:

$$T^* = \frac{E_a}{R \ln(\tau_y^{(0)}/10^6)} \approx \frac{2 \times 10^5}{8.31 \times \ln(10^6)} \approx 1\,800 \text{ K}, \quad (2)$$

where $E_a \approx 150$ – 250 kJ mol^{-1} is the activation energy for silicate flow (Giordano et al., 2008) and R the gas constant. Below T^* , the TMC can no longer eject cohesive material — this threshold constitutes the thermal closure condition of the mechanism.

Rheology 'A $\bar{a}$ — structural property

'A $\bar{a}$ nature of the Coherent Magmatic Torus — intrinsic thermal insulator.

The TMC exhibits a biphasic structure of 'A $\bar{a}$ type: a **mobile plastic core** at high temperature, surrounded by a **fragmented clinker crust** at the surface. This architecture is the direct consequence of the Bingham-Herschel rheology: the outer layer, whose stress is below τ_y , solidifies into solid fragments; the core, hotter and more sheared, remains plastic and mobile.

This structure endows the TMC with a fundamental thermal property: **the clinker crust is an intrinsic thermal insulator**. It blocks radiative energy loss from the plastic core to the exterior, in the same way as a refractory crust in a terrestrial basaltic flow. This insulation is **structural**, not atmospheric: it does not rely on the presence of an external blanketing atmosphere, but on the architecture of the biphasic fluid itself.

Direct consequence: the coherence and persistence of the TMC over the active window (≈ 60 Myr) is not an exceptional duration to be explained by external forcing. It results from the 'A $\bar{a}$ self-thermal insulation, combined with continuous internal heat sources (Fe-Ni segregation, radioactivity, accretion heat) and the maintenance of total melting by the T-Tauri Sun. Cyclic restructuring at each ejection — the crust fragments, the core reorganizes — renews the insulator: the TMC is its own thermal regulator.

Phase transition

First phase transition — rheological. The local crossing of the threshold τ_y is a solid/fluid transition. Below: the TMC is coherent, the magmatic mass resists fragmentation. Above: flow is possible, coherent ejection becomes accessible.

2.3.1 T-Tauri radiative flux as a modulator of τ_y

The yield stress τ_y depends strongly on temperature via an Arrhenius-type relation:

$$\tau_y(T) \approx \tau_y^{(0)} \exp\left(-\frac{E_a}{RT}\right). \quad (3)$$

Over the range $T = 1\,500\text{--}4\,000\text{ K}$, τ_y varies by more than five orders of magnitude. The radiative flux of the T-Tauri Sun maintains the surface in the range $2\,000\text{--}4\,000\text{ K}$:

Key result

The T-Tauri Sun is a direct modulator of the first phase transition. Its accretion outbursts ($L_{\text{outburst}} \approx 3\text{--}10 L_{\odot}$) and its coronal mass ejections (CMEs) keep τ_y in its lowest range ($10^2\text{--}10^4\text{ Pa}$). The active ejection window (4.55–4.49 Ga) *causally* coincides with the T-Tauri phase of the young Sun: as the Sun joins the main sequence and its luminosity stabilizes, τ_y increases and the TMC loses the rheological accessibility that allowed ejections. This is an independent termination mechanism, complementary to the stabilization of obliquity by the growing Moon.

2.4 Oblateness J_2 and shape of the proto-Earth

Rapid rotation ($T_{\text{rot}} = 3.5\text{ h}$) deforms the proto-Earth into an oblate ellipsoid. The dynamic oblateness coefficient J_2 of a rotating fluid body is:

$$J_2 \approx \frac{5}{4} \cdot \frac{\Omega^2 R_{\oplus}}{g_0}, \quad (4)$$

which gives:

$$J_2 \approx \frac{5}{4} \cdot \frac{(4.987 \times 10^{-4})^2 \times 6.371 \times 10^6}{9.81} \approx 0.101. \quad (5)$$

The corresponding equatorial deformation is:

$$\delta R = \frac{J_2}{3} R_{\oplus} \approx 215\text{ km}. \quad (6)$$

The effective radius as a function of the oblique latitude φ is:

$$r(\varphi) = R_{\oplus} \left[1 - J_2 \frac{3 \sin^2 \varphi - 1}{2} \right]. \quad (7)$$

2.5 Silicate atmosphere: confinement, feedback and erosion

The dense silicate atmosphere ($T \approx 2\,000\text{--}4\,000\text{ K}$, $P \approx 10^5\text{--}10^7\text{ Pa}$) plays three active roles: lateral confinement of the TMC during ejection; thermodynamic recalescence of torn-off droplets; and radiative feedback maintaining the surface above the liquidus.

2.5.1 Atmospheric erosion by the T-Tauri wind

The intense ionized wind of the T-Tauri Sun ($\dot{M} \sim 10^{-8}\text{--}10^{-7}\ M_\odot\text{ yr}^{-1}$) erodes the silicate atmosphere from above. The confinement pressure evolves as:

$$P_{\text{conf}}(t) = P_{\text{silicate}}(t) - P_{\text{erosion}}(t), \quad (8)$$

where P_{silicate} is maintained by magmatic degassing and P_{erosion} increases with T-Tauri activity.

Key result

T-Tauri atmospheric erosion facilitates TMC ejection. A less confined atmosphere reduces the vertical barrier opposing the upward motion of TMC material, lowering the effective critical velocity U_{crit} and broadening the ejection window. The temporal evolution of T-Tauri activity provides a **secular drift of the ejection threshold**: the window widens as erosion progresses.

2.6 Chaotic obliquity: fundamental geometric condition

In the absence of a stabilizing satellite, the rotation axis of the proto-Earth is subject to secular resonances with the inner Solar System planets. Simulations by [Laskar et al. \(1993a\)](#); [Laskar & Robutel \(1993b\)](#) show that ε chaotically explores $[0^\circ, 85^\circ]$ over millions of years. The nominal adopted value is $\varepsilon \in [40^\circ, 70^\circ]$, with a central value of 55° .

Key result

Robustness with respect to obliquity. In the range $[40^\circ, 70^\circ]$, the geometric condition $\varepsilon > 45^\circ$ is satisfied about five-sixths of the time. The mechanism thus frequently lifts the Taylor-Proudman (TP) constraint without fine-tuning, exploiting orbital chaos to open recurring ejection windows.

2.7 Stellar environment and early bombardment

The proto-Earth is bathed in an environment dominated by the T-Tauri Sun and an intense planetesimal bombardment. This work treats the T-Tauri environment as an **actively coupled actor** in the triple phase transition, not as a simple background condition.

T-Tauri wind. T-Tauri stars produce an ionized wind with mass-loss rates $\dot{M} \sim 10^{-8}\text{--}10^{-7}\ M_\odot\text{ yr}^{-1}$ and velocities $v_w \sim 300\text{--}500\text{ km s}^{-1}$ ([Feigelson & Montmerle, 1999](#)). The dynamic pressure at $d = 1\text{ AU}$ is:

$$P_w \approx \frac{\dot{M} v_w}{4\pi d^2} \approx 10^{-8}\text{--}10^{-7}\text{ Pa}. \quad (9)$$

This value is at least ten orders of magnitude lower than the dynamic ejection pressure ($\sim 2 \times 10^{11}$ Pa, Section 8): the wind has no direct effect on TMC dynamics. Its indirect roles, however, are substantial.

T-Tauri CME — impulsive trigger of the first episode. Young T-Tauri stars experience frequent coronal eruptions releasing 10^{27} – 10^{29} J (Airapetian et al., 2016), with a frequency that can reach several events per day. Over 10^6 – 10^7 years, the number of opportunities is immense: the probability that no CME occurs during a favourable obliquity window ($\varepsilon > 45^\circ$) is negligible.

Hadean bombardment — injection of turbulent energy. Collisions with bodies of all sizes (residual planetesimals, Grand Tack impactors, Walsh et al. 2011) continuously inject turbulent energy at small scales, feeding the inverse cascade that builds the TMC (Section 4).

Physical note

Three independent contributions of the T-Tauri Sun to the TTP.

1. **Rheological modulator (1st transition):** the T-Tauri radiative flux maintains surface temperatures at 2 000–4 000 K, placing τ_y in its lowest range and making the BH solid→fluid transition accessible at TMC velocities (Section 2.3.1).
2. **Mechanical facilitator (2nd transition):** erosion of the silicate atmosphere reduces the vertical confinement pressure, lowering the effective U_{crit} . CMEs provide finite impulsive perturbations capable of triggering the first episode (Section 2.5.1).
3. **Source of turbulent energy:** the wind, CMEs and diffuse condensates inject energy at small scales, feeding the inverse cascade of the TMC.

The temporal coincidence between the T-Tauri phase (~ 4.55 – 4.50 Ga) and the proposed ejection window (4.55–4.49 Ga) is **causal, not accidental**.

3 Complete force inventory

3.1 Effective gravity — dominant restoring force

The effective gravity at the surface of the oblate rotating proto-Earth is:

$$g_{\text{eff}}(\varphi) = g_0 - \Omega^2 r(\varphi) \sin \varepsilon, \quad (10)$$

where $g_0 = 9.81 \text{ ms}^{-2}$ and $r(\varphi)$ is given by equation (7). At the oblique equator and for $T_{\text{rot}} = 3.5 \text{ h}$: $g_{\text{eff}} \approx 8.51 \text{ ms}^{-2}$, i.e. a reduction of $\approx 13\%$ relative to g_0 .

3.2 Coriolis force — two components, a geometric threshold

For a fluid of velocity U in a rotating frame with angular velocity Ω and obliquity ε , the Coriolis force $\mathbf{F}_C = -2\Omega \times \mathbf{U}$ decomposes into:

$$f_{\text{rad}} = 2\Omega \sin \varepsilon \cdot U, \quad (11)$$

$$f_{\text{horiz}} = 2\Omega \cos \varepsilon \cdot U, \quad (12)$$

with the ratio:

$$\frac{f_{\text{rad}}}{f_{\text{horiz}}} = \tan \varepsilon. \quad (13)$$

For $\varepsilon = 55^\circ$: $\tan 55^\circ \approx 1.43$ — the radial (destabilizing) component dominates the horizontal (confining) component by 43 %.

Key result

The geometric threshold $\varepsilon = 45^\circ$ ($\tan \varepsilon = 1$) is a *necessary* condition for overcoming the Taylor-Proudman (TP) constraint. It follows from the kinematics of the Coriolis force alone, without any adjustable parameter.

3.3 Centrifugal force and oblate deformation

$$F_{cf} = \Omega^2 r \sin \varepsilon. \quad (14)$$

This force concentrates magma in the oblique equatorial bulge, favouring the geometric formation of the TMC.

3.4 Baroclinic pressure gradient — geostrophic organization

In the geostrophic regime (Rossby number $\text{Ro} \ll 1$):

$$\nabla P = -\rho (2\mathbf{\Omega} \times \mathbf{U}), \quad (15)$$

with $\text{Ro} = U/(2\Omega L)$. For the large-scale TMC, $\text{Ro} \approx 1.57 \times 10^{-4} \ll 1$.

3.5 Angular momentum gradient — parameter α

$$\frac{\partial U}{\partial r} \approx -\alpha \frac{U}{r}, \quad (16)$$

where $\alpha < 1$ (sub-rigid), $\alpha = 1$ (Keplerian), $\alpha > 1$ (super-rigid). The status of the hypothesis $\alpha > 1$ is discussed in Section 4.5.

3.6 Solar tidal force — resonant periodic forcing

$$F_{\text{m}\odot} = \frac{2 G M_{\odot} R_{\oplus}}{d_{\odot}^3}. \quad (17)$$

The ratio $F_{\text{m}\odot}/g_{\text{eff}} \approx 10^{-4}$ is too weak to modify λ directly, but the solar tide acts as a resonant periodic forcing at the synodic frequency.

3.7 Proto-lunar tidal force — Mathieu resonance

$$F_{\text{m,Lune}}(t) \propto \frac{M_{\text{Lune}}(t)}{a(t)^3}. \quad (18)$$

This force creates a periodic deformation of the TMC, modifying the radial perturbation equation:

$$\ddot{\xi}_r = [\lambda + A_m(t) \cos(n_{\text{Lune}} t)] \xi_r, \quad (19)$$

which is the Mathieu equation (McLachlan, 1947; Abramowitz & Stegun, 1964). Even for $\lambda < 0$, zones of parametric resonance exist where ξ_r grows exponentially if $n_{\text{Lune}} \approx 2\sqrt{|\lambda|}$ (see subsection 6.6 for a detailed analysis).

3.8 Impacts — impulsive triggers

Grand Tack impactors (Walsh et al., 2011) play three roles: (1) injection of turbulent energy at small scales feeding the inverse cascade; (2) impulsive perturbation of ε ; (3) local overcoming of the threshold τ_y by the shock wave.

3.9 Lorentz force — conditional on the dynamo

$$\mathbf{F}_L = \sigma (\mathbf{U} \times \mathbf{B}) \times \mathbf{B}, \quad (20)$$

with the Elsasser number $\Lambda_{\text{Els}} = \sigma B^2 / (\rho \Omega)$. The dynamo is only detectable from ≈ 4.2 Ga onward (Tarduno et al., 2025), i.e. after the proposed active phase. The Lorentz force thus plays no role during the ejection phase.

3.10 Summary: a metastable system with finite perturbation

Table 3: Force balance and roles in the proposed ejection mechanism. Relative amplitudes are normalized to $g_{\text{eff}} = 1$.

Force	Type	Relative amplitude	am-plitude	Role
Gravity g_{eff}	Static, restoring	1.000		Term c in λ
Radial Coriolis f_{rad}	Destabilizing	~ 0.616		Term b in λ
Gradient $L = rU$	Structural	$\sim 10^{-7}$		Term a in λ
Centrifugal force	Static	~ 0.05		Shapes the TMC
β force	Organizing	~ 0.30		Selects $\ell = 1$
Solar tide	Weak periodic	$\sim 10^{-4}$		Resonant trigger
Proto-lunar tide	Growing periodic	variable		Mathieu resonance
Hadean impacts	Impulsive	$\sim 10^{-2}$		Turbulent injection + trigger
BH rheology τ_y	Plastic threshold	threshold		TMC cohesion
Lorentz force	Magnetic	≈ 0		Negligible (pre-dynamo)
T-Tauri wind	Continuous pressure	$\sim 10^{-15}$		Indirect atmospheric erosion
T-Tauri CME	Impulsive	$\sim 10^{-10}$		Trigger of the first episode

The system is metastable: a finite perturbation — proto-lunar tide in Mathieu resonance, Hadean impact, T-Tauri CME — can tip it into the ejecting state.

4 Geostrophic organization of the TMC

4.1 Geostrophic regime and Taylor-Proudman constraint

In the geostrophic regime ($Ro \ll 1$), the Taylor-Proudman (TP) constraint imposes axial invariance of fluid columns in the direction of $\mathbf{\Omega}$. For $\varepsilon < 45^\circ$, this constraint confines the flow horizontally and forbids radial ejection:

$$Ro_{TMC} = \frac{U_{turb}}{2\Omega R_\oplus} \approx \frac{10^3}{2 \times 4.987 \times 10^{-4} \times 6.371 \times 10^6} \approx 1.57 \times 10^{-4} \ll 1. \quad (21)$$

4.2 Inverse energy cascade

In quasi-2D geostrophic turbulence, energy cascades toward *large* scales (inverse cascade) — the opposite of 3D turbulence (Rhines, 1975; Sukoriansky et al., 2007). This property allows the turbulent energy injected by impactors, convection and the T-Tauri wind to accumulate progressively in the large-scale TMC, without dissipation by a forward cascade.

4.3 Rhines scale and selection of the mode $\ell = 1, m = 1$

The inverse cascade stops at the Rhines scale L_β :

$$L_\beta = \sqrt{\frac{U_{turb}}{\beta}}, \quad (22)$$

where $\beta = 2\Omega \cos \varepsilon / R_\oplus$ is the geophysical beta parameter. For $T_{rot} = 3.5$ h and $\varepsilon = 55^\circ$:

$$\beta = \frac{2 \times 4.987 \times 10^{-4} \times \cos 55^\circ}{6.371 \times 10^6} \approx 8.98 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}, \quad (23)$$

$$L_\beta = \sqrt{\frac{10^3}{8.98 \times 10^{-11}}} \approx 3.34 \times 10^6 \text{ m} \approx 0.52 R_\oplus, \quad (24)$$

corresponding to the spherical harmonic (SH) mode of degree $\ell = 1$, order $m = 1$ — the largest possible planetary mode.

Definition

Coherent Magmatic Torus (TMC). Spatial expression of the dominant mode $\ell = 1, m = 1$ selected by the inverse cascade at the Rhines scale in the zone $|\varphi| < 30^\circ$ of the fully molten proto-Earth in rapid rotation ($Ro \ll 1$). The TMC is not an imposed geometric structure; it is the spatial signature of the energy transfer toward the largest accessible scale in a spherical system in rapid rotation with obliquity. Its coherence is maintained by the Bingham-Herschel (BH) rheology with threshold τ_y , its internal 'Aa'-type structure (intrinsic thermal insulator), and the geostrophic regime.

4.4 Multi-scale architecture

Table 4: Multi-scale architecture of the proposed ejection mechanism.

Phase	Scale	Velocity	Ro	Physics
Organization	$\sim R_{\oplus}$	$\sim 10^3$ m/s	$\sim 10^{-4}$	Geostrophy, inverse cascade
Transition	$L_{\beta} \approx 0.52 R_{\oplus}$	—	~ 1	Turbulence/wave boundary
Ejection	$\sim 10^6$ m	~ 10.4 km/s	~ 4.5	Inertial, hypersonic

4.5 Radial velocity profile: status of the $\alpha > 1$ conjecture

4.5.1 Deformation of Taylor-Proudman columns and super-rotation

The classical picture of Taylor-Proudman columns — perfect cylinders, invariant along the rotation axis — assumes a homogeneous, incompressible, unstratified fluid rotating about a fixed axis. These conditions are not met in the Hadean proto-Earth, for four converging physical reasons:

1. **Strong obliquity and centrifugal force.** For $\varepsilon > 45^\circ$, the centrifugal force ($\Omega^2 r \sin \varepsilon$) deforms the equipotential surfaces and bends the TP columns out of the equatorial plane (Busse, 1970). The restoration of geostrophic equilibrium in this deformed gravity field imposes a more pronounced radial variation of angular momentum than in the canonical Keplerian case.
2. **Density stratification and compositional convection.** The differentiating proto-Earth is not homogeneous: iron sinks, density varies radially. The TP columns are then inclined by buoyancy gradients. Simulations of rapidly rotating convection with stratification show that potential vorticity is redistributed radially, producing intense zonal jets and a super-rotating velocity profile (Aurnou et al., 2007; Gastine et al., 2014).
3. **BH yield-stress rheology.** The TMC is not a Newtonian fluid. Its plasticity allows the TP columns to deform cohesively: the threshold τ_y locally freezes the shear, permitting steeper velocity gradients than in a purely viscous fluid.
4. **Inverse cascade and turbulent forcing.** Energy injection at small scales feeds the inverse cascade toward the $\ell = 1$ mode, sustaining a continuous transfer of angular momentum toward the oblique equator and pushing the system toward a super-rigid regime ($\alpha > 1$) (Sukoriansky et al., 2007).

Physical note

Curvature of TP columns and super-rotation. The combination of strong obliquity, compositional convection and BH rheology constitutes a robust physical mechanism to produce a super-rigid azimuthal velocity profile. In this regime, $\alpha > 1$ is not an arbitrary free parameter: it is an expected signature of the coupled rotation-convection-plasticity dynamics. Scaling laws of planetary super-rotation suggest $\alpha \approx 1.2\text{--}1.5$ (Read et al., 2018).

4.5.2 Central role of α : transition between stability and episodicity

Key result

The system necessarily admits a control parameter α governing the transition between stability and episodicity.

This parameter, defined by $\partial U/\partial r = -\alpha U/r$, controls the sign of the term $a = (1 - \alpha)/r^2$ in the polynomial $\lambda(U) = aU^2 + bU + c$ (equation 35).

- If $\alpha > 1$ ($a < 0$), the parabola is downward-opening: a bounded instability window $[U_1, U_2]$ appears, producing spontaneous episodicity.
- If $\alpha \leq 1$ ($a \geq 0$), the parabola is upward-opening: a single threshold $U > U_{\text{threshold}}$ exists, and episodicity requires external forcing (Mathieu resonance, thermodynamic regulator).

Thus, α is not a mere mathematical convenience: it determines the fundamental dynamic regime of the TMC. Its effective value — which can only be measured by 3D SPH simulation on an oblate sphere (Priority 2 of the validation programme, Section 14) — separates two families of behaviour.

4.5.3 Robustness of the mechanism for $\alpha \leq 1$

The theory remains functional regardless of the α regime:

1. **Super-rigid branch** ($\alpha > 1$): bounded window $[U_1, U_2]$ with $\lambda > 0$, spontaneous episodicity from the structure of λ alone.
2. **Sub-rigid branch** ($\alpha \leq 1$): single instability threshold, episodicity driven by Mathieu resonance and the Ω – R_{Roche} – T thermodynamic regulator.

In both cases, the following elements remain valid: the TMC exists and concentrates energy in the band $|\varphi| < 30^\circ$; the geometric threshold $\varepsilon > 45^\circ$ is necessary; Mathieu resonance triggers successive episodes; the lunar stratigraphy archives progressive differentiation; the predictions on the dynamo and isotopic correlations are independent of α . The parameter α determines the *mechanism* of episodicity, not the existence of lunar formation.

Limitation and open question

Priority 2 of the validation programme. The value of α for the proto-Earth cannot be determined by 2D planar simulation: only a 3D SPH simulation on an oblate sphere is the physically appropriate tool, for three reasons: (1) TP columns are truncated in 2D planar geometry; (2) the parameter β is constant there whereas it varies with latitude on a sphere; (3) the spherical anisotropy of the $\ell = 1$ mode is absent in planar geometry. Measuring α in 3D SPH constitutes Priority 2 of the validation programme (Section 14), consecutive to seismological data (Priority 1).

4.5.4 Angular momentum relaxation time

The derivation of λ assumes that the displaced parcel conserves its angular momentum $H_3 = rU$ during displacement. Above the threshold τ_y , the effective viscosity drops: the relaxation time of angular momentum by viscous diffusion is:

$$\tau_{\text{relax}} \sim \frac{\rho \ell^2}{\eta_{\text{eff}}}, \quad (25)$$

where $\ell \approx 10^6 \text{ m}$ is the parcel scale and $\eta_{\text{eff}} = k \dot{\gamma}^{n-1} \approx 10^4 \text{ Pa s}$ the effective dynamic viscosity. One obtains:

$$\tau_{\text{relax}} \approx \frac{3\,300 \times (10^6)^2}{10^4} \approx 3 \times 10^{11} \text{ s}, \quad (26)$$

to be compared with the dynamic time $\tau_{\text{dyn}} \sim \ell/U \approx 10^6/10^4 \approx 10^2 \text{ s}$. The ratio $\tau_{\text{relax}}/\tau_{\text{dyn}} \approx 3 \times 10^9 \gg 1$: the parcel conserves its angular momentum over the entire duration of the dynamic displacement. The hypothesis of H_3 conservation is therefore physically justified.

5 Necessary conditions for the mechanical transition

5.1 Geometric threshold $\varepsilon > 45^\circ$

For $\varepsilon > 45^\circ$, the radial Coriolis component dominates the horizontal component: the Taylor-Proudman (TP) constraint is lifted and radial ejection becomes geometrically possible (Section 3).

5.2 Reduction of effective gravity

For $T_{\text{rot}} = 3.5 \text{ h}$ and $\varepsilon = 55^\circ$:

$$g_{\text{eff}} = 9.81 - (4.987 \times 10^{-4})^2 \times 6.371 \times 10^6 \times \sin 55^\circ \approx 8.51 \text{ m s}^{-2}, \quad (27)$$

i.e. a reduction of $\approx 13\%$ relative to g_0 .

5.3 Critical ejection velocity and cancellation of the Coriolis deficit

$$U_{\text{crit}} = \sqrt{2 g_{\text{eff}} r(0)} = \sqrt{2 \times 8.51 \times 6.371 \times 10^6} \approx 10.41 \text{ km s}^{-1}. \quad (28)$$

We define the dimensionless parameter:

$$b' = \frac{2\Omega \sin \varepsilon \cdot U_{\text{crit}}}{|g_{\text{eff}}|}, \quad (29)$$

which measures the ratio of the radial Coriolis force to the effective gravity at the ejection threshold. For the nominal values:

$$b' = \frac{2 \times 4.987 \times 10^{-4} \times \sin 55^\circ \times 10\,410}{8.51} \approx 1.000. \quad (30)$$

Key result

Remarkable property at $T_{\text{rot}} = 3.5 \text{ h}$. The destabilizing Coriolis term exactly compensates the effective gravity at the ejection threshold. The system is marginally unstable with respect to an infinitesimal perturbation: ejection is accessible without a mandatory finite perturbation for the first episode. This setting emerges directly from the equations — it is not postulated — and anchors the value of T_{rot} in the dynamics of the model.

5.3.1 Tolerance to obliquity variations

For $\varepsilon = 50^\circ$: $b' \approx 0.93$ (deficit of 7%). For $\varepsilon = 45^\circ$: $b' \approx 0.83$ (deficit of 17%). These margins are surmountable by finite perturbations (CMEs, impacts, solar tide). The mechanism exploits a wide and recurring window, not a single critical value.

Table 5: Necessary conditions for the radial transition.

Condition	Expression	Nominal value	Status
Obliquity $> 45^\circ$	$\varepsilon > 45^\circ$	55° (range 40° – 70°)	Necessary
TP lifted	$\tan \varepsilon > 1$	1.43	Necessary
Critical velocity	$U \geq U_{\text{crit}}$	≈ 10.41 km/s	Necessary
Coriolis compensation	$b' \approx 1.000$	≈ 1.000	Favourable

5.4 Role of the deficit at $T_{\text{rot}} > 3.5$ h

For $T_{\text{rot}} = 5.5$ h, the parameter $b' \approx 0.62$: the Coriolis force compensated only 62% of the effective gravity, leaving a deficit of 38%. The system was formally stable in a linear analysis, imposing an obligatory triggering by a finite perturbation. This deficit does not invalidate the theory; it indicates that ultra-rapid rotation is a necessary condition for the TMC to reach the threshold $\lambda > 0$ without external input.

5.5 Ejection criterion by finite perturbation — potential barrier

$$\delta E > \Delta E_{\text{barr}} = \int_{U_{\text{TMC}}}^{U_1} |\lambda(U)| dU. \quad (31)$$

6 Second phase transition: the TMC and episodic ejection

6.1 Energy accumulation in the TMC — recharge time

Kinetic energy accumulates progressively in the TMC between two ejection episodes. The recharge time τ_{cd} depends on temperature via the Arrhenius law:

$$\tau_{\text{cd}}(T) \propto \exp\left(\frac{E_a}{k_B T}\right), \quad (32)$$

where E_a is the activation energy of magmatic viscosity and k_B the Boltzmann constant. Between 3500 K and 2000 K, τ_{cd} increases by a factor $\sim 30\,000$. This exponential growth contributes to the increasing spacing of late episodes, reinforcing the bimodality prediction (Prediction P4, Section 13) and anchoring the thermodynamic regulator in a secular trend.

6.2 Stability coefficient $\lambda(U, \varepsilon, \alpha)$

The evolution of an infinitesimal radial perturbation ξ_r of the TMC is governed by:

$$\ddot{\xi}_r = \lambda \xi_r. \quad (33)$$

This coefficient is derived in the fluid parcel displacement formalism — analogous to the Solberg-Høiland criterion for a homogeneous rotating fluid with obliquity (Solberg, 1936; Høiland, 1941). We recall here its final form, while the complete derivation is presented below (Section 6.5).

The algebraic sum of the three physical contributions is:

$$\lambda = \underbrace{g_{\text{eff}}}_{\text{eff. gravity (stabilizing)}} + \underbrace{2\Omega \sin \varepsilon \cdot U}_{\text{radial Coriolis (destabilizing)}} + \underbrace{\frac{U^2}{r^2}(1 - \alpha)}_{\text{gradient of } L=rU \text{ (Rayleigh criterion, Rayleigh 1917)}} \quad (34)$$

For $\lambda > 0$: instability (ejection); for $\lambda < 0$: stability (confinement).

Condition for the existence of the ejection mechanism

Key result

If $(\Omega, \varepsilon, \alpha, \tau_y, \dots)$ satisfy the following conditions:

- $\varepsilon > 45^\circ$ (lifting of the Taylor-Proudman constraint),
- τ_y sufficiently low so that $\Pi = \rho U^2 / \tau_y \gg \Pi_{\text{crit}}$ (sheet cohesion),
- U in the instability window (either $U > U_{\text{threshold}}$ if $\alpha \leq 1$, or $U_1 < U < U_2$ if $\alpha > 1$),
- and (for episodes 2 to N) the Mathieu resonance condition $n_{\text{Lune}} \approx 2\sqrt{|\lambda|}$,

$\implies$ **then the ejection mechanism exists.**

6.3 Polynomial structure and two physical regimes

Setting $a = (1 - \alpha)/r^2$, $b = 2\Omega \sin \varepsilon > 0$, $c = g_{\text{eff}} < 0$:

$$\lambda(U) = aU^2 + bU + c. \quad (35)$$

The sign of a , determined by α , governs the structure of the instability.

6.3.1 Sub-rigid or iso-rigid regime ($\alpha \leq 1$)

Then $a \geq 0$: the polynomial $\lambda(U)$ is an upward-opening parabola (or a straight line if $\alpha = 1$). The condition $\lambda > 0$ defines a single threshold $U > U_{\text{threshold}}$. Episodicity cannot emerge spontaneously from the structure of λ alone; it must be driven by an external mechanism (Mathieu resonance, thermodynamic regulator).

6.3.2 Super-rigid regime ($\alpha > 1$)

Then $a < 0$: the polynomial $\lambda(U)$ is a downward-opening parabola. With $c = g_{\text{eff}} < 0$, the condition $\lambda > 0$ is satisfied only inside a bounded interval $U_1 < U < U_2$ (Proposition 6.1). The system then exhibits natural episodicity: every ejection removes the TMC from the window, imposing a recharge time τ_{cd} before the next episode.

Key result

Robustness of the mechanism regardless of α . Whether α is greater or less than 1:

- the TMC exists and concentrates energy in the band $|\varphi| < 30^\circ$,
- the geometric threshold $\varepsilon > 45^\circ$ is necessary for lifting the TP constraint,
- Mathieu resonance can trigger multiple ejections,
- the lunar stratigraphy archives progressive differentiation,
- the predictions on the dynamo and isotopic correlations are independent of α .

The parameter α determines the **mechanism** of episodicity, not the existence of lunar formation.

6.4 Bounded instability window (case $\alpha > 1$)

Proposition 6.1 (Bounded instability window). *For $\alpha > 1$, $a < 0$: the parabola $\lambda(U)$ is concave downwards. With $c = g_{\text{eff}} < 0$, it can be positive only inside a bounded interval $U_1 < U < U_2$. Setting $\beta_\alpha = \alpha - 1 > 0$:*

$$U_{1,2} = \frac{r^2}{\beta_\alpha} \left[\Omega \sin \varepsilon \mp \sqrt{\Omega^2 \sin^2 \varepsilon - \frac{\beta_\alpha}{r^2} |g_{\text{eff}}|} \right]. \quad (36)$$

This window exists if and only if:

$$\Delta = 4 \left(\Omega^2 \sin^2 \varepsilon - \frac{\beta_\alpha}{r^2} |g_{\text{eff}}| \right) > 0, \quad (37)$$

which imposes a dynamic threshold on obliquity:

$$\varepsilon > \varepsilon_0 = \arcsin \left(\frac{\sqrt{\beta_\alpha |g_{\text{eff}}|}}{r \Omega} \right). \quad (38)$$

Corollary 6.1 (Double threshold). *The mechanism is controlled by two thresholds:*

1. *Geometric threshold $\varepsilon = 45^\circ$: pure Coriolis kinematics, with no adjustable parameter.*
2. *Dynamic threshold ε_0 : existence of the bounded window $[U_1, U_2]$ (only if $\alpha > 1$).*

Activation requires $\varepsilon > \max(45^\circ, \varepsilon_0)$.

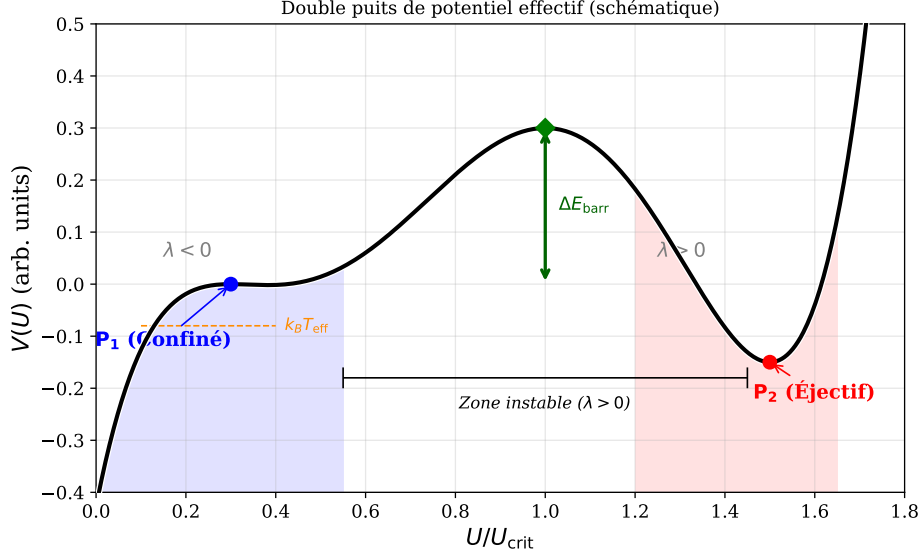


Figure 1: Double well of effective potential $V(U)$ of the TMC. P_1 : confined state ($\lambda < 0$), accumulation by τ_{cd} . P_2 : ejecting state ($\lambda > 0$), capture f_{cap} . The barrier ΔE_{barr} is overcome by available finite perturbations: pressure waves from planetesimal impacts, CMEs from the T-Tauri Sun, Mathieu resonance with the growing proto-lunar tide. The level $k_B T_{eff}$ represents the effective turbulent agitation. The barrier lowers at each episode k due to the decrease of $g_{eff}^{(k)}$. © Michel Debaillleul — ULB 2026.

6.5 Complete derivation of $\lambda(U, \varepsilon, \alpha)$

6.5.1 Preamble: the fluid parcel displacement formalism

This subsection constitutes the analytical foundation of the second phase transition. It details the derivation of λ in the fluid parcel displacement formalism — the classical analogue of the Solberg-Høiland criterion for a homogeneous rotating fluid with obliquity (Solberg, 1936; Høiland, 1941).

Consider a fluid parcel of the TMC at radial distance r , with azimuthal velocity $U(r)$ in the frame rotating at Ω . It is subjected to a virtual radial displacement $\xi_r > 0$ and the net radial force is calculated. The linearized form of this balance to first order in ξ_r leads to $\ddot{\xi}_r = \lambda \xi_r$.

6.5.2 Underlying hypotheses

H1: Rapid displacement relative to viscous diffusion. $\tau_{dyn} \sim r/U_{crit} \approx 10^3 \text{ s} \ll \tau_\nu \sim r^2/\nu_{eff}$.

H2: Anelastic hypothesis. Pressure adjusts instantaneously around the displaced parcel.

H3: Local conservation of angular momentum. The parcel conserves $L = rU(r)$ during the displacement.

6.5.3 The three contributions to λ

Gravitational contribution (stabilizing).

$$\lambda_1 = g_{eff} < 0. \quad (39)$$

Radial Coriolis contribution (destabilizing).

$$\Omega = \Omega \cos \varepsilon \hat{z}_{rot} + \Omega \sin \varepsilon \hat{r}_{eq}, \quad (40)$$

$$\lambda_2 = +2\Omega \sin \varepsilon \cdot U > 0. \quad (41)$$

Contribution of radial shear — Rayleigh criterion.

$$U_{\text{parcel}} \approx U(r) \left(1 - \frac{\xi_r}{r} \right), \quad (42)$$

$$U_{\text{background}}(r + \xi_r) \approx U(r) \left(1 - \alpha \frac{\xi_r}{r} \right), \quad (43)$$

$$\lambda_3 = \frac{U^2}{r^2} (1 - \alpha), \quad (44)$$

the generalized Rayleigh criterion (Rayleigh, 1917).

6.5.4 Final expression of λ

$$\lambda(U, \varepsilon, \alpha) = \underbrace{g_{\text{eff}}}_{\text{eff. gravity}} + \underbrace{2\Omega \sin \varepsilon \cdot U}_{\text{radial Coriolis}} + \underbrace{\frac{U^2}{r^2} (1 - \alpha)}_{\text{gradient of } L=rU} \quad (45)$$

in polynomial form in U :

$$\lambda(U) = aU^2 + bU + c, \quad a = \frac{1 - \alpha}{r^2}, \quad b = 2\Omega \sin \varepsilon, \quad c = g_{\text{eff}}. \quad (46)$$

6.5.5 Remarkable property at $T_{\text{rot}} = 3.5 h$

$$b' = \frac{2\Omega \sin \varepsilon \cdot U_{\text{crit}}}{|g_{\text{eff}}|} = \frac{2 \times 4.987 \times 10^{-4} \times 0.8192 \times 10\,410}{8.51} \approx 1.000. \quad (47)$$

6.5.6 Scope and limitations of λ

Key result

The coefficient λ provides a *necessary* condition for the radial instability of the TMC.

Limitation and open question

The parcel displacement formalism is a linear, local and instantaneous approximation. Validation is delegated to 3D SPH simulations (Priorities 3 and 4).

6.6 Mathieu resonance as a trigger for episodes 2 to N

The Mathieu equation (19) admits unstable solutions in frequency bands centred on $n_{\text{Lune}} \approx 2\sqrt{|\lambda|}/k$ for $k = 1, 2, 3, \dots$. Even for $\lambda < 0$, the growing proto-lunar tide can trigger instability through parametric resonance.

6.6.1 Setting the problem

$$A_m(t) \approx \frac{2 G M_{\text{Lune}}(t)}{a(t)^3} R_{\oplus}, \quad (48)$$

leading to the forced perturbation equation:

$$\ddot{\xi}_r = [\lambda + A_m(t) \cos(n_{\text{Lune}} t)] \xi_r. \quad (49)$$

6.6.2 Canonical Mathieu form

$$\frac{d^2 \xi_r}{d\theta^2} + (a - 2q \cos 2\theta) \xi_r = 0, \quad (50)$$

with (McLachlan, 1947; Abramowitz & Stegun, 1964):

$$a = -\frac{4\lambda}{n_{\text{Lune}}^2}, \quad q = \frac{2A_m}{n_{\text{Lune}}^2}. \quad (51)$$

6.6.3 Instability zones and resonance condition

$$n_{\text{Lune}} \approx 2\sqrt{|\lambda|}. \quad (52)$$

$$\mu \approx \frac{1}{2} \sqrt{q^2 - (a - 1)^2}. \quad (53)$$

$$\mu \frac{n_{\text{Lune}}}{2} \gtrsim \frac{1}{\tau_{\text{cd}}}. \quad (54)$$

Key result

Mathieu resonance provides a physically founded mechanism for triggering episodes 2 to N , even when $\lambda < 0$.

6.7 Initiation of the first ejection episode

Mathieu resonance presupposes an Orbital Seeding Body (CAO, defined in Section 7) already in orbit. This condition is satisfied for episodes 2 to N , but it cannot explain episode 1. The fully molten proto-Earth, in rapid rotation ($T_{\text{rot}} = 3.5$ h) and at high obliquity, is an intrinsically metastable system. The first episode results from a **cooperative triggering** of hierarchical mechanisms:

1. **Rapid rotation — dominant force.** Rotation at $T_{\text{rot}} = 3.5$ h is the main source of kinetic energy. Without it, no other mechanism can operate.
2. **Direct rotational instability.** The combination of the centrifugal force at the oblique equator, the BH yield-stress rheology and chaotic obliquity may be sufficient to extract the first plastic sheet.
3. **Solar tide.** The tide of the T-Tauri Sun provides a resonant periodic forcing at the synodic frequency.

4. **Impact triggering.** An impactor of a few tens to hundreds of kilometres, frequent in the Hadean environment (Walsh et al., 2011), can locally overcome the threshold τ_y and initiate plastic extraction.
5. **T-Tauri CME — finite impulsive perturbation.** Coronal mass ejections (10^{27} – 10^{29} J, Airapetian et al. 2016), with a frequency of several events per day, provide an immense number of opportunities over the active window.
6. **Thermal convection.** Vigorous convection continuously injects turbulent energy at small scales, feeding the inverse cascade.

Limitation and open question

The triggering of the first episode is the least constrained part of the mechanism. Unlike the following episodes, for which Mathieu resonance provides a precise analytical framework, the initial initiation relies on cooperative triggering whose quantitative efficiency is not calculated in this work. 3D SPH simulation will have to test whether this coupling effectively produces a first body in orbit of sufficient mass to initiate the feedback loop.

Phase transition

Second phase transition — mechanical. The TMC is a cosmic relaxation oscillator: it accumulates energy (metastable state), then tips into the ejecting state under the effect of a resonant finite perturbation. Each tipping produces an ejection episode and forms an additional layer on the Orbital Seeding Body (Section 7). Episodicity emerges either from the bounded window $[U_1, U_2]$ for $\alpha > 1$, or from the coupling between Mathieu resonance and the thermodynamic regulator for $\alpha \leq 1$.

6.8 Hierarchical causal chain and self-consistent feedback loop

Key result

The physical regimes of the TTP are not a mere juxtaposition of independent conditions; they form a causal hierarchy in which each step engenders and stabilizes the next.

The logic of the mechanism can be made explicit as a cascade of deterministic constraints:

1. **Rapid rotation** ($T_{\text{rot}} = 3.5 \text{ h}$) $\implies \text{Ro} \ll 1 \implies$ **geostrophic regime.**
2. **Geostrophy** $\implies$ quasi-2D turbulence $\implies$ **inverse energy cascade** (Rhines, 1975; Sukoriansky et al., 2007).
3. **Inverse cascade + β effect** $\implies$ selection of the **mode** $\ell = 1$ at the Rhines scale $L_\beta \approx 0.52 R_\oplus$.
4. **Mode $\ell = 1$ + BH rheology** $\implies$ **TMC**, a self-insulating coherent structure of 'A $\bar{a}$ ' type.
5. **TMC** $\implies$ **accumulation of kinetic energy** in the band $|\varphi| < 30^\circ$.
6. **Accumulation + BH threshold exceeded** $\implies$ **coherent hypersonic ejection.**
7. **Ejection** $\implies$ **depletion of the mantle in siderophiles + rotational slowing Ω .**

Physical note

Analogy with the stellar main sequence. A star on the main sequence simultaneously satisfies hydrostatic equilibrium, radiative transport, nuclear fusion, convection and a magnetic field. These regimes result from a causal hierarchy initiated by gravitational collapse. The Hadean proto-Earth is, in an analogous way, a physical object whose regimes are hierarchically determined by rapid rotation and Fe-Ni differentiation.

7 The Orbital Seeding Body**Posture note**

This section presents the Orbital Seeding Body (CAO), a central architectural piece of the theory. The CAO is the first object placed in stable orbit during the first ejection episode. It constitutes the gravitational core onto which the sheets ejected during the following six episodes accumulate by concentric plastic accretion, forming the Moon through progressive construction.

7.1 Definition and physical parameters of the CAO**Definition**

Orbital Seeding Body (CAO). First solid-plastic object placed in stable orbit around the proto-Earth during the first TMC ejection episode. The CAO constitutes the unique accretion centre onto which the sheets of episodes 2 to N are deposited by plastic welding. It originates from the differentiating Hadean silicate mantle, of isotopic composition identical to that of the contemporary proto-Earth.

7.1.1 Mass and dimensions

The mass ejected per episode is estimated at $M_{\text{ej}} \approx M_{\text{Lune}}/N \approx M_{\text{Lune}}/7$. For the first episode:

$$d_{\text{CAO}} \approx 160 \text{ km}, \quad r_{\text{CAO}} = 80 \text{ km}. \quad (55)$$

The corresponding mass for a density $\rho \approx 3\,300 \text{ kg m}^{-3}$:

$$M_{\text{CAO}} = \frac{4}{3}\pi r_{\text{CAO}}^3 \rho \approx \frac{4}{3}\pi \times (8 \times 10^4)^3 \times 3\,300 \approx 7.1 \times 10^{18} \text{ kg}. \quad (56)$$

7.1.2 Initial orbital distance

$$d_{\text{CAO}} \approx 3\text{--}6 R_{\oplus} \approx 19\,000\text{--}38\,000 \text{ km}. \quad (57)$$

This distance is beyond the Roche radius ($R_{\text{Roche}} \approx 3.6 R_{\oplus} \approx 23\,000 \text{ km}$) for the most energetic episodes.

7.1.3 Cohesion of the sheet: parameter Π

$$\Pi = \frac{\rho U_{\text{crit}}^2}{\tau_y} \approx \frac{3\,300 \times (10\,410)^2}{10^3} \approx 3.57 \times 10^8 \gg \Pi_{\text{crit}} \approx 10^{-2}. \quad (58)$$

7.1.4 Isotopic composition of the CAO: mechanical guarantee

The plastic sheet extracted from the TMC is taken from the Hadean silicate mantle at the moment of ejection. Its isotopic composition in oxygen, silicon and titanium is identical to that of the contemporary proto-Earth, without an exogenous source.

Key result

Earth-Moon isotopic similarity: direct mechanical consequence. Prediction P1 ($\Delta^{17}\text{O} < 5$ ppm Earth-Moon) follows mechanically from the fact that the lunar material is Hadean terrestrial mantle contemporary with each ejection. There is no exogenous isotopic source.

7.1.5 Iron depletion: driver of lunar stratigraphy

At each successive episode, Fe-Ni differentiation has progressed: the mantle source of the TMC is increasingly depleted in iron. This results in a natural concentric stratigraphy:

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k+1}, \quad k = 1, \dots, N-1. \quad (59)$$

The inner layers (early episodes) are richer in iron; the outer layers (late episodes) are more depleted. This radial Fe/Si gradient is Prediction P6.

7.2 Intrinsic rotation of the CAO and concentric plastic accretion

7.2.1 Intrinsic rotation of the CAO: inheritance from the TMC

The CAO originates from the rapidly rotating TMC ($\Omega \approx 4.987 \times 10^{-4} \text{ rad/s}$). By conservation of angular momentum during plastic consolidation in orbit, the CAO inherits a significant intrinsic rotation.

7.2.2 Enrobing mechanism: the sheet wraps around

Each subsequent episode sends a cohesive plastic sheet toward the CAO. This sheet encounters a rotating body that **wraps it around itself** by centrifugal effect and viscoplastic deformation:

1. The sheet leaves the TMC at $U \approx U_{\text{crit}} \approx 10.41 \text{ km s}^{-1}$.
2. It deforms plastically during its transit.
3. It reaches the sphere of influence of the CAO and wraps progressively by concentric plastic welding.
4. The mass and radius of the CAO increase by discrete steps.

Physical note

The intrinsic rotation of the CAO solves the impact problem. The intrinsic rotation of the CAO transforms impact into enrobing: the sheet arrives on a body already moving in the same direction. The relative velocity is absorbed gradually, layer after layer.

7.2.3 *The Moon: an onion-like stratigraphic archive***Key result**

Internal architecture: seven concentric layers. The inner layers (episodes 1–3) correspond to the iron-richest Hadean mantle (little-advanced differentiation). The intermediate layers (episodes 4–5) carry the geochemical transition. The outer layers (episodes 6–7) correspond to the most siderophile-depleted mantle.

7.2.4 *Growth of the lunar radius per episode*

$$R_{\text{Lune}}(k) = R_{\text{CAO}} \left(1 + \sum_{j=1}^{k-1} \frac{f_{\text{cap}} M_{\text{ej}}^{(j)}}{M_{\text{CAO}}} \right)^{1/3}. \quad (60)$$

7.3 The proto-Earth as a variable-mass spinning top

$$M_{\oplus}^{(k)} = M_{\oplus}^{(0)} - \sum_{j=1}^k M_{\text{ej}}^{(j)}, \quad (61)$$

$$I_{\text{sys}}^{(k)} = \frac{2}{5} M_{\oplus}^{(k)} R_{\oplus}^2 \left(1 - \frac{2}{5} J_2^{(k)} \right), \quad (62)$$

$$\delta\Omega_{\text{ejection}}^{(k)} = - \frac{M_{\text{ej}}^{(k)} U_{\text{crit}} R_{\text{TMC}}}{I_{\text{sys}}^{(k)}}, \quad (63)$$

$$\Omega^{(k+1)} = \Omega^{(k)} + \delta\Omega_{\text{ejection}}^{(k)} + \delta\Omega_{\text{tide}}^{(k)}. \quad (64)$$

7.4 Uniqueness of the Moon: four converging arguments

Argument 1 — The CAO is unique by construction. The TMC, organized by the spherical harmonic mode $\ell = 1$, $m = 1$, produces a single preferential ejection direction.

Argument 2 — Mathieu resonance can sustain only one oscillator. A second body in a different orbit would create a non-resonant forcing frequency. The first body captures the resonant monopole.

Argument 3 — Concentric plastic accretion is a unique attractor. The sphere of influence of the CAO, growing at each episode, captures the flux before it can reconsolidate into an independent object.

Argument 4 — The thermodynamic regulator extinguishes the mechanism. Extinction is irreversible once ε is stabilized at 23.5° by the growing Moon.

8 Coherent hypersonic ejection

8.1 Constitutive nature of Hadean magma and cohesion at high velocity

The partially crystallized Hadean silicate magma is not a Newtonian fluid. It is a Bingham-Herschel (BH) yield-stress fluid, and this property is **constitutive**: it follows from the thermal state, pressure and composition of the medium at that time.

Dynamic ejection pressure.

$$P_{\text{dyn}} = \frac{1}{2} \rho U_{\text{crit}}^2 \approx 1.78 \times 10^{11} \text{ Pa} \approx 178 \text{ GPa}. \quad (65)$$

Mach number. The speed of sound in the biphasic fluid ($\phi \approx 0.2$): $c_s \approx 0.9 \text{ km s}^{-1}$ (Ayukai & Kanagawa, 2025; Carrara et al., 2022; Mader et al., 2013), hence:

$$\text{Ma} = \frac{U_{\text{crit}}}{c_s} \approx \frac{10.41}{0.9} \approx 11.6. \quad (66)$$

Physical note

The non-Newtonian rheology is not a hypothesis: it is what this material is. The dynamic pressure $\Pi \approx 3.57 \times 10^8$ exceeds the BH rupture threshold by eight orders of magnitude: the sheet is intrinsically cohesive.

8.2 The three closure conditions of the model

$$\eta_{\text{TMC}} E_{\text{rot}} \geq E_{\text{esc}}, \quad (67)$$

$$\Pi = \frac{\rho U_{\text{crit}}^2}{\tau_y} \gtrsim \Pi_{\text{crit}} \approx 10^{-2}, \quad (68)$$

$$f_{\text{cap}} \gtrsim \frac{v_\infty^2}{2 U_{\text{crit}}^2}. \quad (69)$$

9 Non-ballistic orbital capture and stratigraphic construction of the Moon

9.1 Analytical model of non-ballistic capture

$$r_H = a \left(\frac{M_p}{3 M_\oplus} \right)^{1/3}, \quad (70)$$

$$\Theta_g = \frac{G M_p}{R_p U^2}, \quad (71)$$

$$r_{\text{infl}} = r_H (1 + \Theta_g), \quad (72)$$

$$\sigma_{\text{eff}} = \pi r_{\text{infl}}^2, \quad (73)$$

$$f_{\text{geo}} = \frac{\sigma_{\text{eff}}}{L_{\text{nappe}}^2}, \quad (74)$$

$$\Pi_{\text{cap}} = \frac{\tau r_{\text{infl}}}{\rho U^2 R_p}, \quad (75)$$

$$f_{\text{cap}} \approx \frac{\pi r_H^2 (1 + \Theta_g)^2}{L_{\text{nappe}}^2} \left[1 - \exp\left(-\frac{\tau r_H (1 + \Theta_g)}{\rho U^2 R_p}\right) \right] \quad (76)$$

Limitation and open question

Status of the capture model. Equation (76) is an analytical parametrization, not a simulation result. The 3D SPH simulation remains the ultimate arbiter on the value of f_{cap} (Prediction P12, Section 14).

9.2 The Moon as a concentric stratigraphic archive

Radial Fe/Si gradient.

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k+1}, \quad k = 1, \dots, N-1. \quad (77)$$

The lunar magma ocean does not erase the deep stratification. The formation of a Lunar Magma Ocean (LMO) after final consolidation is probable. This ocean is estimated at a maximum depth of 300–500 km (Elkins-Tanton et al., 2011). The layers of the early episodes reside at greater depths: they are not affected by the thermal mixing of the LMO.

9.3 Three-way correlation Fe/Si — Hf-W — dynamo

$$\frac{d}{dt} \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{Lune}} \propto \frac{dM_{\text{core}}}{dt} \propto B_{\text{Hadean}}(t). \quad (78)$$

Falsifiable prediction

Prediction P20. The mass fraction ejected per episode increases monotonically from episode 1 to episode N . The quantitative target is $M_{\text{ej}}(N)/M_{\text{ej}}(1) \gtrsim 2$.

10 Episodic dynamics and chronology

10.1 The thermodynamic regulator Ω – R_{Roche} – T

The extinction of the mechanism is emergent, resulting from the convergence of three physically independent decays:

$$\Lambda_{\text{eff}}^*(t) = f_{\Omega}(t) \cdot f_{R_{\text{Roche}}}(t) \cdot f_T(t) \xrightarrow{t \rightarrow t^*} 0. \quad (79)$$

$$f_{\Omega}(t) = \left(\frac{\Omega(t)}{\Omega_0} \right)^2 = \left(1 - \frac{t}{\tau_{\Omega}} \right)^2, \quad (80)$$

$$f_{R_{\text{Roche}}}(t) \propto M_{\text{Lune}}^{-2/3}(t), \quad (81)$$

$$f_T(t) = \exp\left(-\frac{\tau_{\text{cd}}(t)}{\tau_{\text{cd}}(0)}\right) \approx \exp\left(-A e^{B/T(t)}\right). \quad (82)$$

10.2 Role of tides: rotational slowing and obliquity stabilization

Rotational slowing by the lunar tide. The tide exerted by the growing Moon gradually transfers the terrestrial rotational angular momentum to the orbital angular momentum. From $T_{\text{rot}} = 3.5$ h at formation, the Earth progressively reaches the current 24 h over 4.5 Gyr.

Stabilization of obliquity at 23.5° . The Moon, through its spin-orbit resonance, irreversibly locks $\varepsilon \approx 23.5^\circ$ as soon as its mass reaches $M_{\text{Lune}} \approx M_{\oplus}/80$ (Laskar et al., 1993a).

Key result

Direct consequences: two verified predictions. Rotational slowing and the stabilization of obliquity at 23.5° are *direct consequences* of the formation of the Moon by the proposed mechanism.

10.3 Thermal consequence of obliquity stabilization

Key result

Axis stabilization and accelerated cooling. When the growing Moon stabilizes the terrestrial obliquity around 23.5° , the chaotic mechanical heating associated with axis flips ceases abruptly. This accelerated surface cooling leads to the formation of a solid proto-crust and to the appearance of liquid water as early as ~ 4.4 Ga, consistent with the oldest Jack Hills zircons.

10.4 Triangulation of the number of episodes $N \approx 7 \pm 2$

Table 6: Triangulation of N by three independent routes.

Constraint	Reasoning	N
Mass budget	$N = M_{\text{Lune}}/\langle M_{\text{ej}} \rangle$	~ 7
Angular momentum	Conservation $L_{\oplus} \rightarrow L_{\text{orb}}$	~ 6
Time window	$N = \tau_{\text{active}}/\langle \tau_{\text{cycle}} \rangle$	~ 8
Convergence	Triangulation	7 ± 2

10.5 Bimodality: early chaotic phase and late regulated phase

The temporal distribution of episodes is bimodal: 3–4 close episodes ($< 30\,000$ yr) in the initial chaotic phase, then 3–4 increasingly spaced episodes when the Ω – R_{Roche} – T regulator becomes active (Prediction P4).

10.6 Extinction of the mechanism

Extinction is irreversible: once ε is stabilized at 23.5° , $\tan \varepsilon \approx 0.43 < 1$ — the condition $\varepsilon > 45^\circ$ is never satisfied again.

11 Third phase transition: the progressive terrestrial dynamo

11.1 Growth of the Fe-Ni core — source of convection

$$M_{\text{core}}(t) = M_{\text{core}}^{\infty} \left(1 - e^{-t/\tau_{\text{seg}}} \right), \quad (83)$$

$$Q_{\text{conv}}(t) \approx \Delta g_{\text{Fe}} \frac{M_{\text{core}}^{\infty}}{\tau_{\text{seg}}} e^{-t/\tau_{\text{seg}}}. \quad (84)$$

11.2 Elsasser number and detectability threshold

$$\Lambda_{\text{Els}}^{\text{crit}} \sim 10^{-1} - 10^0. \quad (85)$$

$$B_{\text{crit}} = \sqrt{\frac{\Lambda_{\text{Els}}^{\text{crit}} \rho \Omega}{\sigma}} \approx 6 \times 10^{-4} \text{ T} \approx 6 \mu\text{T}. \quad (86)$$

Phase transition

Third phase transition — magnetic. The dynamo emerges gradually as compositional convection in the growing Fe-Ni core exceeds $\Lambda_{\text{Els}}^{\text{crit}}$. It is not the *appearance* of the dynamo that is dated at 4.2 Ga: it is its *crossing of the detectability threshold*.

11.3 Observational constraint: Jack Hills zircons

Single-crystal paleointensity analyses by Tarduno et al. (2015, 2020, 2025) establish a robust geomagnetic field at ≈ 4.2 Ga. No robust signal is documented earlier. This timing is a direct prediction of the theory (Prediction P17).

12 Coupling of the three transitions and lunar internal stratification

12.1 Fe-Ni differentiation as the unique driver

- **Synchronous coupling (transitions 1 and 2).** Differentiation depletes the mantle in siderophiles, simultaneously modifying the TMC rheology and the energy available for ejections.
- **Delayed consequence (transition 3).** Successive ejections accelerate differentiation. The dynamo emerges as a detectable signal only ≈ 300 Myr after the extinction of ejections.

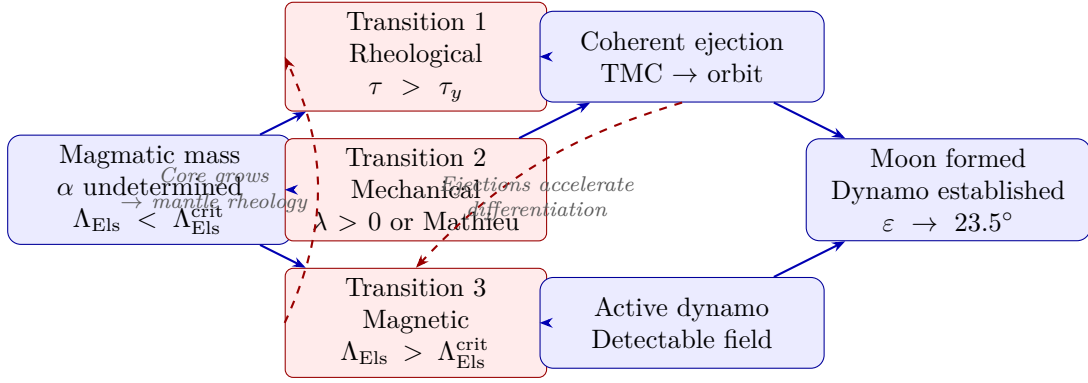


Figure 2: Phase diagram of the proto-Earth system.

12.2 Irreversible self-extinction

The mechanism is self-extinguishing by construction. Once ε is stabilized at 23.5° , the threshold $\varepsilon > 45^\circ$ is never satisfied again.

12.3 Lunar internal stratification: foundations, partial preservation and seismological programme

Posture note

This section develops the central structural consequence of concentric plastic accretion: the Moon should exhibit an internal stratification in successive layers, each carrying the geochemical imprint of the Earth's mantle at the precise moment of its ejection. It then proposes a seismological observation programme as the priority validation pathway, prior to and independent of any 3D numerical simulation.

12.3.1 The onion architecture: direct mechanical consequence

Non-ballistic plastic accretion by concentric enrobing (Section 7.2) implies that the Moon was built by the addition of successive sheets around the CAO. Since each sheet came from the terrestrial mantle at a different stage of Fe-Ni differentiation, this construction mechanically generates a **radial stratification in iron composition**:

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer } k+1}, \quad k = 1, \dots, N-1, \quad (87)$$

the inner layers being the richest in iron (little-differentiated Hadean mantle, episodes 1–3), the outer layers the most depleted (advanced Hadean mantle, episodes 5–7). This radial Fe/Si gradient constitutes Prediction P6.

A density contrast is associated with this compositional gradient. For a silicate magma, density varies approximately as:

$$\Delta\rho_{k,k+1} \approx \left[\left(\frac{\text{Fe}}{\text{Si}}\right)_k - \left(\frac{\text{Fe}}{\text{Si}}\right)_{k+1} \right] \times \Delta\rho_{\text{Fe/Si}}, \quad (88)$$

where $\Delta\rho_{\text{Fe/Si}}$ is the mass density contrast between ferrous olivine and magnesian olivine, of the order of $100\text{--}200\text{ kg m}^{-3}$ for a Mg# variation of about 10 units (Garcia et al., 2019).

This density contrast between successive layers constitutes a potentially detectable seismic interface.

12.3.2 Estimation of interface depths: two scenarios

The depth of the interfaces between layers depends on the distribution of masses ejected per episode. Prediction P20 posits $M_{\text{ej}}(N)/M_{\text{ej}}(1) \gtrsim 2$, without fixing the ratio q of the progression. Two scenarios are presented to evaluate the robustness of the estimate.

Scenario A — equal masses per episode ($q = 1$). If each episode ejects an identical mass $M_{\text{ej}}^{(k)} = M_{\text{Lune}}/N$ for $N = 7$, the radius after k layers is:

$$R_k^{(A)} = R_{\text{Lune}} \left(\frac{k}{N} \right)^{1/3}, \quad (89)$$

and the depth of the interface between layers k and $k + 1$ from the lunar surface is:

$$d_k^{(A)} = R_{\text{Lune}} - R_k^{(A)} = R_{\text{Lune}} \left[1 - \left(\frac{k}{N} \right)^{1/3} \right]. \quad (90)$$

Table 7: Scenario A — Equal masses ($q = 1$, $N = 7$). $R_{\text{Lune}} = 1737$ km, mean density $\bar{\rho} = 3346 \text{ kg m}^{-3}$.

Episode k	Radius R_k (km)	Depth d_k (km)	Zone
1 (CAO)	893	844	Deep mantle
2	1125	612	Deep mantle
3	1284	453	Preservation window
4	1416	321	Preservation window
5	1531	206	Preservation window
6	1635	102	Upper mantle
7	1737	0	Surface

Scenario B — increasing masses ($q > 1$, $M_{\text{ej}}(7)/M_{\text{ej}}(1) = 2$). If the ejected masses follow a geometric progression of ratio q :

$$M_{\text{ej}}^{(k)} = M_{\text{ej}}^{(1)} \cdot q^{k-1}, \quad (91)$$

with the constraint $\sum_{k=1}^7 M_{\text{ej}}^{(k)} = M_{\text{Lune}}$ and $M_{\text{ej}}(7)/M_{\text{ej}}(1) = 2$ (lower bound of P20), we obtain $q \approx 1.122$ and $M_{\text{ej}}^{(1)} \approx 0.087 M_{\text{Lune}}$.

The radius after k layers is:

$$R_k^{(B)} = R_{\text{Lune}} \left(\frac{\sum_{j=1}^k M_{\text{ej}}^{(j)}}{M_{\text{Lune}}} \right)^{1/3}, \quad (92)$$

$$d_k^{(B)} = R_{\text{Lune}} - R_k^{(B)}. \quad (93)$$

Table 8: Scenario B — Increasing masses ($q \approx 1.122$, $M_{\text{ej}}(7)/M_{\text{ej}}(1) = 2$, $N = 7$).

Episode k	Radius R_k (km)	Depth d_k (km)	Zone
1 (CAO)	800	937	Deep mantle
2	1034	703	Deep mantle
3	1210	527	Preservation window
4	1366	371	Preservation window
5	1506	231	Preservation window
6	1626	111	Upper mantle
7	1737	0	Surface

Key result

Robustness of the preservation window. In both scenarios, the interfaces of layers 3–4–5 lie between ≈ 200 and ≈ 530 km depth. This convergence is a structural property of the model: it is independent of the precise value of q in the range allowed by P20. The window 200–530 km constitutes the priority target of the proposed seismological programme.

12.3.3 Two destructive processes and a preservation window

The onion architecture is not necessarily intact today. Two independent processes may have altered it:

From below — deep perturbation. Residual heat from differentiation, deep mantle convection, and the overturn of ilmenite cumulates (Briaud et al., 2023) may have remelted or homogenized the deepest layers — episodes 1 and 2, residing at more than 600–900 km depth depending on the scenario. The ilmenite-enriched low-velocity zone (LVZ), identified seismically and geodetically (Briaud et al., 2023; Weber et al., 2011), testifies to this deep mobility. These inner layers carry the signature of the iron-richest Hadean mantle; their possible homogenization does not erase the global compositional gradient but reduces the stratigraphic resolution.

From above — superficial perturbation. Late volcanism of the LMO, followed by mare basalt volcanism active until ≈ 2 Ga (Li et al., 2021; Che et al., 2021), has covered, remelted or disrupted the most superficial layers — episodes 6 and 7, residing at less than 100–110 km depth. The original stratigraphy of these layers is probably erased or masked.

The possible preservation window. Between these two processes, the middle layers — episodes 3, 4 and 5, residing between ≈ 200 and ≈ 530 km depth — may have escaped both alterations: too deep for surface volcanism, too shallow for deep cumulate overturn.

Limitation and open question

Status of this estimate. The existence of a partial preservation window is a *possible* consequence of the theory, not a certainty. It rests on order-of-magnitude estimates of alteration depths. Lunar mantle convection could have mixed these layers over 4.5 Gyr. Only high-resolution seismic data will be able to decide.

12.3.4 State of the art in lunar seismology and geodesy

Lunar seismology has known three successive generations of investigation.

Apollo data (1969–1977). The four seismometers deployed at Apollo 12, 14, 15 and 16 sites — all on the nearside — recorded hundreds of deep moonquakes (800–1100 km), shallow moonquakes (50–200 km) and impacts. These data established a large-scale model of the internal structure: a bipartite crust (20 km upper + 40–55 km lower depending on site), a mantle with quasi-homogeneous velocity down to ≈ 1100 km, and a deep attenuation zone.

A crucial result for the present theory is that *the Apollo data are entirely compatible with a homogeneous mantle without large radial velocity variations* (Khan et al., 2006; Garcia et al., 2019): no mid-mantle discontinuity is *required* by these data. However, their insufficient resolution and the limited coverage of the nearside alone do not allow exclusion of such a stratification.

Reprocessing and geodetic contributions (2011–2023). Weber et al. (2011) reanalysed Apollo seismograms by array methods, identifying: a solid inner core, a liquid outer core, and a partially molten layer at the core-mantle boundary. Briaud et al. (2023) combined geophysical and geodetic constraints to demonstrate that only models incorporating a **low-viscosity zone (LVZ) enriched in ilmenite** and an inner core exhibit densities compatible with the measured tidal deformations.

12.3.5 A seismological programme as a priority validation pathway

Key result

Hierarchy of the proposed validation programme. Lunar seismology constitutes **Priority 1** of the validation programme: it provides direct observational constraints on the internal structure, at lower cost and in a shorter timeframe than 3D SPH simulations. The 3D numerical simulation (Priority 2 and later) will be all the better constrained if it has seismic data to calibrate its input parameters.

Farside Seismic Suite (FSS). First seismometer ever deployed on the lunar farside, it carries two seismometers from the InSight mission (VBB + SP). FSS will land in Schrödinger basin, offering a relatively thin crust — a condition favourable for wave propagation toward the deep mantle (Panning et al., 2022).

Lunar Environment Monitoring Station (LEMS). Compact and autonomous seismometer, planned to be deployed by Artemis III astronauts in the south polar region, with a sensitivity increased by an order of magnitude relative to Apollo instruments (LEMS, 2024).

The combination of FSS (farside) + LEMS (south pole) will for the first time offer sufficient angular coverage for tomographic inversions of the lunar mantle at intermediate resolution.

Observables expected to test Prediction P22. Phase conversions at ferrous-composition interfaces in the 200–530 km window; lateral velocity variations consistent with the $m = 1$ mode (Prediction P7); a radial stratification of the attenuation factor Q .

12.4 Synthesis: a favourable time window

The set of available data — partially constrained lunar internal structure, imminent seismological missions, persistence of geochemical debates on mantle composition — outlines a favourable time window in which prediction P6/P22 of this theory can be tested in a direct and independent manner. This window is on the order of the decade 2025–2035, making it one of the most immediately testable predictions of the present work.

13 Falsifiable predictions

Table 9: Twenty-three falsifiable predictions. **A:** testable now. **B:** medium term. **C:** simulation or future mission.

Level	#	Prediction and causal chain
A	P1	$\Delta^{17}\text{O} < 5$ ppm Earth-Moon (common source: upper Hadean mantle)
A	P2	Lunar siderophile depletion $\propto$ segregation depth of the terrestrial core
A	P3	Non-linear Hf-W chronology (discrete episodes)
A	P4	Bimodality of inter-episode intervals
A	P5	Absence of a geochemical tracer of an external impactor in the Moon
B	P6	Fe/Si gradient increasing with lunar depth
B	P7	$m = 1$ zonal structure of the lunar mantle
B	P8	Lunar mass $\propto \Omega^2$
B	P9	Lunar D/H compatible with a terrestrial magmatic source
B	P10	Residual $v_z \sim 10\text{--}20 \text{ m s}^{-1}$ at the current orbit
C	P11	SPH: cohesion of the TMC flux at $\text{Ma} \approx 11.6$
C	P12	SPH: capture fraction f_{cap}
C	P13	SPH: measurement of α on an oblate sphere
C	P14	SPH: geostrophic $\rightarrow$ ejection transition
C	P15	SPH: surface $\lambda = 0$ in (ε, U, α)
C	P16	SPH: dependence of f_{cap} on oblate geometry
A/B	P17	<i>Gradual</i> growth of the Hadean field over 4.55–4.2 Ga
B/C	P18	Lunar Fe/Si — paleointensity (Tarduno et al., 2025) — Hf-W chronology correlation
C	P19	Apollo zircons: TRM consistent with a sub-critical terrestrial field
B	P20	Ejected mass per episode increasing, $M_{\text{ej}}(N)/M_{\text{ej}}(1) \gtrsim 2$
B	P21	Peripheral micro-droplets: bimodal intercalations
B/C	P22	Lunar seismic tomography: impedance contrasts at layer interfaces between ≈ 200 and ≈ 530 km depth, testable by FSS, LEMS and the lunar seismic network under construction (Panning et al., 2022; LEMS, 2024)
C	P23	Chemistry/age correlation of Hadean zircons at resolution $< 10^5$ yr

14 Validation programme

Key result

Hierarchy of the validation programme. The programme follows a natural scientific logic: observation first, modelling afterwards. Lunar seismological data (Priority 1) will provide the observational constraints necessary for calibrating numerical simulations (Priorities 2–5). This sequence is not a convenience: submitting a theory to the observational arbiter before numerical simulation is the approach that allowed characterization of the terrestrial internal structure.

14.1 Priority 1 — Lunar seismological network

Expected observables to test against Prediction P22: (1) phase conversions at ferrous-composition interfaces in the 200–530 km window; (2) lateral velocity variations consistent with the $m = 1$ mode (Prediction P7); (3) radial stratification of the attenuation factor Q .

14.2 Priorities 2 to 5 — Numerical simulations

- **Priority 2:** Measurement of α on an oblate sphere by 3D spherical geostrophic turbulence simulation (P13).
- **Priority 3:** Geostrophic $\rightarrow$ ejection transition in 3D SPH on an oblate sphere, $\varepsilon > 45^\circ$, BH fluid (P14).
- **Priority 4:** Hypersonic cohesion with BH threshold in compressible SPH, $\text{Ma} \approx 11.6$ (P11, P12).
- **Priority 5:** Coupled differentiation–dynamo model (P17, P18).

A complete algorithmic skeleton for the 3D SPH simulation (Priority 2) is provided in an annex to this document, including diagnostics for measuring α , λ , TMC energy and the capture fraction. The estimated computational resources (≈ 28 h on 4 A100 GPUs) are compatible with standard access to national high-performance computing infrastructures.

15 Discussion

15.1 Responses to objections

Response to an objection

Objection 1: “The TMC is not demonstrated.” The TMC is not arbitrarily postulated: it is the spatial expression of the dominant mode $\ell = 1$, $m = 1$ selected by the inverse cascade at the Rhines scale in a spherical system in rapid rotation with obliquity greater than 45° . This emergence is founded on established physical mechanisms (Rhines, 1975; Sukoriansky et al., 2007). The quantitative validation is the object of Priority 2 of the validation programme (3D SPH on an oblate sphere).

Response to an objection

Objection 2: “Cohesion at $\text{Ma} \approx 12$ is physically improbable.” The sheet cohesion is a consequence of the constitutive nature of Hadean magma (BH law). Jet stability is characterized by an inertio-cohesive regime: $\Pi \sim 10^8$, $We \gg 1$, $\rho_{\text{flux}}/\rho_{\text{atm}} \approx 200$. The relevant criterion is $Bi_{KH} \gg 1$, not We . Final validation is entrusted to Priority 4 of the SPH programme.

Response to an objection

Objection 3: “ $\alpha > 1$ is not demonstrated.” The theory does not depend on $\alpha > 1$ for its existence, but for the *mechanism* of episodicity. In both branches, the Moon forms by concentric plastic accretion, and the 23 predictions remain valid.

Response to an objection

Objection 4: “The regimes are too constrained to coexist.” These regimes are not independent; they form a causal hierarchy (Section 6.8). Each step engenders the next, from rapid rotation to ejection, in a manner analogous to the nesting of regimes in a main-sequence star.

Response to an objection

Objection 5: “A 3D SPH simulation is necessary for the theory to be acceptable.” The 3D SPH simulation is necessary to validate the internal dynamic mechanisms, but it is neither the only validation pathway, nor the first. Lunar seismology directly tests the most observable consequence of the theory — the internal stratification — without resorting to numerical modelling. Proposing a fully coupled 3D SPH simulation as a prerequisite for the receivability of a theory would be to confuse definitive validation with scientific receivability.

15.2 Honest limitations

1. **Status of α .** The value of α for the $\ell = 1$ mode in oblate spherical geometry cannot be measured by 2D planar simulation — only 3D spherical SPH is the appropriate tool.
2. **Atmospheric pressure in λ .** The addition of a pressure gradient $\partial P / \partial r < 0$ is not quantified.
3. **Geostrophic-ejection transition not simulated.** The λ analysis is a necessary, not sufficient, condition.
4. **Hypersonic regime with BH threshold.** The $\text{Ma} \approx 11.6$ regime for dense plastic materials at high temperature is physically poorly constrained.
5. **Initial eccentricity of the CAO.** Circularization is not quantified.
6. **Capture fraction f_{cap} .** The analytical model is a parametrization to be validated.
7. **Partial melting during plastic welding.** Not modelled.
8. **Volatiles in the rheology.** Unquantified systematic uncertainty.
9. **Preservation state of the middle layers.** The estimate of the 200–530 km window rests on order-of-magnitude assumptions on alteration depths. Mantle convection over 4.5 Gyr could have partially homogenized these layers.

These limitations define the current frontiers of the theory and its forthcoming scientific programme.

16 Conclusion

16.1 Synthesis of the triple phase transition

This work proposes that the formation of the Moon is not a single catastrophic event but the by-product of a triple phase transition in a fully molten proto-Earth undergoing continuous differentiation. The unique driver — progressive segregation of iron toward the core — simultaneously produces three ruptures of physical regime: rheological, mechanical and magnetic. These three transitions are coupled and hierarchically ordered, each preparing the conditions for the next.

16.2 What the theory demonstrates, proposes and predicts

The theory *demonstrates*: the geometric condition ($\varepsilon > 45^\circ$), the polynomial structure of $\lambda(U, \varepsilon, \alpha)$, and spontaneous episodicity for $\alpha > 1$.

It *proposes*: the finite-perturbation ejection criterion via Mathieu resonance, the continuous description of the BH plastic flux of 'A $\bar{a}$ ' type, the three-way correlation (lunar Fe/Si — Hf-W — paleointensity), and the concentric internal lunar architecture in seven layers as a direct mechanical consequence of non-ballistic plastic accretion.

It *predicts*: twenty-three falsifiable observables (Table 9), including the radial stratification in iron composition (P6) and seismic impedance contrasts in the 200–530 km window (P22).

16.3 Lunar stratification: mechanical consequence and observational target

The onion structure of the Moon — seven concentric layers each carrying the geochemical imprint of the Earth's mantle at the moment of its ejection — is one of the most directly testable consequences of the theory. It is not postulated: it follows mechanically from non-ballistic plastic accretion. It is also not in contradiction with existing data: Apollo seismological observations are compatible with a homogeneous mantle, but their insufficient resolution does not allow exclusion of a stratification at the proposed scale.

Two processes could have partially altered this architecture: the overturn of deep ilmenite cumulates (Briaud et al., 2023) from below, and late volcanism until 2 Ga (Li et al., 2021; Che et al., 2021) from above. These two processes define a **possible preservation window** for the middle layers (episodes 3–4–5), between ≈ 200 and ≈ 530 km depth.

16.4 A sequenced validation programme

1. **Lunar seismological data** (Priority 1, horizon 2025–2035): FSS, LEMS (Artemis III), Chinese missions, ESA ECLIPSE network. These data will directly test P22 and constrain the parameters of subsequent simulations.
2. **3D SPH simulation on an oblate sphere** (Priority 2): measurement of α and validation of the geostrophic-ejection transition.
3. **Coupled differentiation–dynamo simulations** (Priorities 3–5): hypersonic cohesion, Mathieu resonance, evolution of the Hadean dynamo.

16.5 Open questions

Several fundamental questions remain open and constitute the natural research agenda of this work: the value of α in oblate spherical geometry; the capture fraction f_{cap} in a non-ballistic plastic accretion regime; the actual preservation state of the middle layers after 4.5 Gyr of lunar thermal evolution; the quantitative relationship between the lunar Fe/Si gradient and

measurable seismic velocities; the exact nature of the first ejection episode and the minimum mass of the CAO to initiate the feedback loop.

These questions are not weaknesses of the theory: they define its current frontiers and its forthcoming scientific programme.

16.6 Popular summary

4.55 billion years ago, the Earth is a ball of completely molten rock spinning on itself in 3.5 hours. Its axis wobbles chaotically — the Moon does not yet exist to stabilize it — while iron and nickel, being heavier, slowly sink toward the centre to form the core.

At the equator of this oblate Earth, rotational forces organize the magma into an immense annular belt of 'A $\bar{a}$ ' type — the Coherent Magmatic Torus — racing at high speed in a band of $\pm 30^\circ$. This belt progressively accumulates energy, like a spring being compressed. Its clinker crust thermally insulates it from the exterior: it persists for 60 million years without dissolving.

The first time, a combination of impacts, rotational instability and coronal outbursts from the young Sun suffices to trigger an ejection: material is hurled into space at more than $40\,000\text{ km h}^{-1}$ and captured in orbit. Henceforth, the young forming Moon exerts a resonant gravitational push on the torus, triggering subsequent episodes. This phenomenon occurs about seven times over 40 to 60 million years.

The Moon grows by successive wrapping of plastic sheets around the first orbital body. Each sheet comes from the terrestrial mantle at a different stage of its differentiation: the first sheets are the richest in iron, the last the poorest. The Moon is thus built like an onion, layer after layer, archiving in its internal structure the history of the birth of the Earth's core.

Its tidal force gradually slows the Earth's rotation — from 3.5 hours to 24 hours — and definitively stabilizes its axis at 23.5° . The iron sinking toward the core feeds an increasingly vigorous convection. About 300 million years after the end of ejections — about 4.2 billion years ago — this motion becomes intense enough to generate a coherent magnetic field.

The deepest layers of this lunar onion were probably remelted or mixed over time by internal heat. The most superficial layers were covered by volcanism. But the intermediate layers, between 200 and 530 kilometres depth, may have survived, still preserving today the trace of seven birth episodes. The seismographs that Artemis III astronauts will deploy on the Moon in the coming years — and those that robotic probes will place on the farside — may have the necessary sensitivity to hear these layers resonate.

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